RetinaFace-R50
Face Detection Performance Report

DeepStream + TensorRT FP16 · GPU Blur Pipeline

Work Item INTER-69 · Face Detection + GPU Blur

MEASURED REPORT INTER-69-RetinaFace-R50 · October 2026

Report ID	INTER-69-RetinaFace-R50
Work Item	INTER-69 · Face Detection + GPU Blur
Model	RetinaFace-R50
Task	Face detection + GPU blur
Device	NVIDIA GeForce RTX 5080 (16 GB)
Runtime	DeepStream + TensorRT (FP16)
Input size	1280 × 736
Script	retinaface_face_detection.py
Status	Measured · performance evidence recorded

Prepared in the layout and control style of the xis.ai AI Departmental Handbook v1.0.

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1 Executive Summary

Purpose. This report records measured performance for **RetinaFace-R50** in the INTER-69 face detection + GPU blur pipeline on NVIDIA GeForce RTX 5080.

Measured finding. RetinaFace-R50 detects the most faces (77,585 total, 43.10 avg per frame) at 346.1 FPS. It uses the most GPU (74.6%) and has higher per-stage latency due to the larger 1280×736 input.

1.1 Common settings

Setting	Value
Device	NVIDIA GeForce RTX 5080 (16 GB)
Runtime	DeepStream + TensorRT (FP16)
Model	RetinaFace-R50
Detection size	1280 × 736
Confidence / NMS	0.40 / 0.4
Frames	1800 (warmup 30 excluded from timings)
Frames dropped	0
Bottleneck	encode (NVENC)

1.2 Summary metrics (video file, 1920×1080)

Metric	Value
Avg Pipeline FPS	346.1
Estimated Max FPS	452.21
Headroom	1.13×
Total Detections	77,585
Avg Faces / Frame	43.10
Max Faces / Frame	55

1.3 Latency per frame (ms)

Stage	mean	max
Detect (nvinfer / TensorRT)	1.82	6.14
Post (decode + NMS)	0.80	2.42
Blur (CUDA kernels)	1.36	5.81
Encode (NVENC)	2.21	4.68

Note: Detect and Blur rows highlighted in green. Total Detections highlighted in yellow.

1.4 Resource usage

Metric	Value
CPU cores used (avg)	1.56
GPU utilisation avg / peak	74.6% / 77%
GPU memory peak	4398 MB

2 Module Execution Placement

Module	Device
Stream Decode (NVDEC)	GPU

Face Detection Model (TensorRT)	GPU
Post-processing (decode + NMS)	CPU (lightweight)
Face Blur (Custom CUDA kernels)	GPU
Video Encode (NVENC)	GPU
Stream Mux / Demux / Queue (DeepStream)	GPU

Almost the entire pipeline runs on GPU. Only the tiny detector output tensor decoding + NMS runs on CPU.

4 Key Takeaways

• **RetinaFace-R50** achieved **346.1 FPS** on a 1920×1080 video file with 0 frames dropped.

• Total detections: **77,585** (avg 43.10 faces/frame, max 55).

• GPU utilisation avg / peak: **74.6% / 77%**; memory peak: **4398 MB**.

• Real live streams (cameras / Rutube) are limited by source rate (~25–30 FPS). The pipeline itself is much faster.